

CARBOXYLATED NBR PATENT LANDSCAPE: DIRECT COPOLYMERIZATION AND PROCESS-CONTROLLED LATEX TERMINATION

1. ABSTRACT

This report synthesizes two primary patents relevant to carboxylated nitrile rubber (carboxylated NBR) within the 2010–2024 US/EP search scope. The target material is defined by incorporation of carboxyl functionality into the nitrile-butadiene rubber backbone, typically through copolymerization of acrylonitrile and butadiene with an unsaturated carboxylic acid monomer such as methacrylic acid or acrylic acid. The selected patents show two distinct but complementary technology directions: one patent directly discloses a carboxylated NBR synthesis route, while the other is a non-carboxylated styrene-butadiene copolymer patent that is only tangentially related and serves mainly as a contrast case for acid-monomer-free hot polymerization.

US20200283557A1 is the key primary patent for the target compound. It discloses emulsion/latex polymerization of acrylonitrile, 1,3-butadiene, and methacrylic acid in water, using t-dodecyl mercaptan as chain-transfer agent, cumene hydroperoxide as initiator, and a pressure reactor held at 5 °C. The inventive focus is not the carboxylation chemistry itself, but the termination strategy: a nitrous acid salt combined with a water-insoluble hydroquinone is added at a controlled conversion to suppress discoloration and stabilize the latex and final rubber. The examples map the effect of sodium nitrite dosage and hydroquinone co-terminator level on pH, conversion drift, Mooney viscosity drift, discoloration, and scorch stability.

US9815984B2 does not disclose carboxylated NBR synthesis; instead, it teaches hot polymerization of styrene-butadiene copolymers in the absence of acid monomers, followed by sulfur curing and use in asphalt systems. It is included because it provides a useful boundary condition for the search space: hot polymerization without carboxyl monomers is explicitly non-target, but its process parameters and latex handling illustrate how acid-free rubber dispersions are managed in related elastomer technology.

Overall, the patent landscape indicates that the most relevant synthesis route for carboxylated NBR is direct free-radical emulsion polymerization of butadiene/acrylonitrile with a carboxylic acid comonomer, followed by carefully controlled polymerization termination and coagulation. The strongest evidence supports methacrylic-acid-containing formulations, low-temperature initiation

around 5 °C, and low-dose nitrite/hydroquinone termination to minimize discoloration while preserving storage stability.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US20200283557A1
- Patent Title: Method for producing carboxylated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses emulsion/latex polymerization of a monomer mixture containing an α,β -ethylenically unsaturated nitrile monomer and a carboxyl group-containing monomer in water, followed by termination of the polymerization with a nitrous acid salt plus a water-insoluble hydroquinone and coagulation of the latex to isolate the solid rubber. The inventive emphasis is on controlling discoloration and storage stability of the carboxylated nitrile rubber by using a specific terminator combination at low dosage. -

1,3-Butadiene: 68 parts — Example 1 - Acrylonitrile: 25 parts — Example 1 - Methacrylic acid: 7 parts — Example 1 - t-Dodecyl mercaptan: 0.5 parts — Example 1 - Deionized water: 200 parts — Example 1 - Sodium dodecylbenzenesulfonate: 2.0 parts — Example 1 - Sodium β -naphthalenesulfonate formalin condensate: 1.0 part — Example 1 - Cumene hydroperoxide: 0.15 parts — Example 1 - Polymerization temperature: 5 °C — Example 1 - Polymerization conversion ratio at termination: 65% — Example 1 - Sodium nitrite: 0.025 parts — Example 1 - 2,5-di-t-amylhydroquinone dispersion: 1.6 parts of 25 wt % aqueous dispersion (0.4 parts active) — Example 1 - pH of latex: 4.2 — Example 1 - 1,3-Butadiene: 68 parts — Example 6 - Acrylonitrile: 25 parts — Example 6 - Methacrylic acid: 7 parts — Example 6 - t-Dodecyl mercaptan: 0.5 parts — Example 6 - Deionized water: 200 parts — Example 6 - Sodium dodecylbenzenesulfonate: 2.0 parts — Example 6 - Sodium β -naphthalenesulfonate formalin condensate: 1.0 part — Example 6 - Cumene hydroperoxide: 0.15 parts — Example 6 - Polymerization temperature: 5 °C — Example 6 - Polymerization conversion ratio at termination: 80% — Example 6 - Sodium nitrite: 0.05 parts — Example 6 - 2,5-di-t-amylhydroquinone dispersion: 0.8 parts of 25 wt % aqueous dispersion (0.2 parts active) — Example 6 - pH of latex: 5.5 — Example 6 - Acrylonitrile: 25 parts — Table 1 / Examples 1-7 - 1,3-Butadiene: 68 parts — Table 1 / Examples 1-7 - Methacrylic acid: 7 parts — Table 1 / Examples 1-7 - Sodium nitrite: 0.025 to 0.2 parts — Table 1 / Examples 1-7 - 2,5-di-t-amylhydroquinone: 0.1 to 0.4 parts active —

Table 1 / Examples 1-7 - Hydroxylamine sulfate: 0.2 parts — Comparative Example 1 - Diethylhydroxyamine: 0.2 parts — Comparative Example 2 - Methylhydroquinone: 0.2 parts — Comparative Example 3 - Sodium nitrite: 0.05 parts — Comparative Example 5 - Sodium nitrite: 0.2 parts — Comparative Example 6 - Sodium nitrite: 0.2 parts — Comparative Example 7 - 2,5-di-t-amylhydroquinone: 0.2 parts active — Comparative Example 7

Relevant Experimental Evidence - Example 1: Demonstrated successful preparation of carboxylated nitrile rubber latex at 5 °C using acrylonitrile/butadiene/methacrylic acid and nitrite plus water-insoluble hydroquinone termination. - Example 2: Showed that increasing sodium nitrite to 0.05 parts still gave a stable latex with low discoloration. - Example 3: Showed that adding more hydroquinone dispersion with nitrite maintained good termination and latex quality. - Example 4: Showed that 0.15 parts sodium nitrite remained within the effective range for low discoloration and stability. - Example 6: Demonstrated termination at higher conversion (80%) while still obtaining a stable carboxylated nitrile rubber latex.

Technical Relevance This is the most directly relevant patent in the set because it explicitly discloses synthesis of carboxyl group-containing nitrile rubber from acrylonitrile, butadiene, and methacrylic acid. It also identifies a practical process window for termination chemistry that improves color and storage stability, which is highly relevant to manufacturable carboxylated NBR.

Patent 2

- Patent Number: US9815984B2
- Patent Title: Non-carboxylated styrene-butadiene copolymers, preparation method and use ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hot polymerization of styrene and butadiene in aqueous medium at 40 °C or greater to produce a non-carboxylated styrene-butadiene copolymer, followed by sulfur curing and use in asphalt-modified compositions. The process explicitly excludes acid monomers, so it is not a direct route to carboxylated NBR, but it provides a related high-temperature latex polymerization framework. - Polymerization temperature: 40 °C or greater — Claim 1 / synthesis passages - Polymerization temperature: 50 °C or greater — synthesis passages - Weight ratio of styrene to butadiene: 20:80 to 80:20 — Claim 1 / synthesis passages - Weight ratio of styrene to butadiene: 25:75 — synthesis passages - Cis-1,4 butadiene units: greater than 20% — Claim 1 / synthesis passages - Trans-1,4 butadiene units: less than 60% — Claim 1 / synthesis passages - Styrene-butadiene copolymer amount: 0.5% to 30% based on total solids — Claim 5 - Blending temperature: 150 °C or greater — synthesis passages - Polymer A: 3 wt % — Example 1 / Example 6 / Table 4 - Polymer B: 3 wt % — Example 3 / Example 5 / Table 4 - Polymer C: 3 wt % — Example 1 / Example 2 / Example 4 /

Table 2 - Curing agent: 2.1 wt % — Example 2 / Example 3 / Example 4 - Cationic surfactant: 4 parts — Example 4 - Non-ionic surfactant: 2 parts — Example 4

Relevant Experimental Evidence - Example 1: Showed asphalt viscosity differences between Polymer A and Polymer C modified samples. - Example 2: Showed the effect of 2.1 wt % curing agent on Polymer C-modified asphalt. - Example 3: Showed Polymer B-modified asphalt viscosity and the effect of curing agent. - Example 4: Showed elastic recovery and sweep performance of asphalt emulsions containing hot-polymerized SBR dispersions. - Example 7: Showed that Polymer B produced substantially lower asphalt viscosity than Polymer A or Polymer A plus curing agent.

Technical Relevance This patent is only indirectly relevant because it explicitly teaches non-carboxylated, acid-monomer-free rubber synthesis. It is useful as a contrast reference for process conditions in related latex polymerization, but it does not advance carboxylated NBR chemistry.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US20200283557A1 is a direct carboxylated NBR patent; US9815984B2 is explicitly non-carboxylated and therefore outside the target chemistry.
- US20200283557A1 uses acrylonitrile/butadiene/methacrylic acid monomers, whereas US9815984B2 uses styrene/butadiene only and excludes acid monomers.
- US20200283557A1 is a low-temperature emulsion polymerization at 5 °C with chain-transfer agent and peroxide/redox initiation; US9815984B2 teaches hot polymerization at 40 °C or greater, with some embodiments at 50 °C or greater.
- US20200283557A1 emphasizes polymerization termination control using sodium nitrite plus water-insoluble hydroquinone; US9815984B2 emphasizes sulfur curing and asphalt blending rather than termination chemistry.
- US20200283557A1 reports conversion-dependent termination and latex pH values around 3.8 to 5.6; US9815984B2 reports formulation and performance metrics for asphalt blends, not monomer conversion or coagulation conditions.

4. CONCLUSION

The evidence supports direct free-radical emulsion polymerization of acrylonitrile, butadiene, and a carboxylic acid comonomer, with methacrylic acid clearly demonstrated, as the most relevant route to carboxylated NBR. The strongest process lesson from the target patent is that low-temperature polymerization at 5 °C combined with controlled termination using a nitrous acid salt and a water-insoluble hydroquinone can suppress discoloration and improve storage stability. Based on the disclosed examples, sodium nitrite at 0.025 to 0.15 parts per 100 parts monomer, together with

0.1 to 1.0 part hydroquinone on a monomer basis, is the most evidence-backed termination window. The second patent is not a synthesis route for carboxylated NBR, but it reinforces that acid-free hot polymerization belongs to a different technology class. For carboxylated NBR manufacture, the recommended evidence-based starting point is acrylonitrile/butadiene/methacrylic acid emulsion polymerization at about 5 °C with chain-transfer control and nitrite/hydroquinone termination before coagulation.

5. REFERENCES

- US20200283557A1 | Method for producing carboxylated nitrile rubber | Not disclosed | US | 2020 | <https://patents.google.com/patent/US20200283557A1/en>
- US9815984B2 | Non-carboxylated styrene-butadiene copolymers, preparation method and use ... | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9815984B2/en>